Pinakeshwar Vaishnav

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PROFESSIONAL PROFILE

Highly analytical, results-driven **AI Engineering & Full-Stack Professional** with a proven track record of architecting and deploying end-to-end, high-performance intelligent systems. Completely self-taught with deep first-principles expertise across Machine Learning mathematics, multi-modal generative AI pipelines, and robust software engineering. Adept at breaking down complex problems, optimizing high-throughput model serving infrastructure, and shipping production-ready code. Demonstrated capability to excel in fast-paced, remote environments, successfully transitioning advanced AI research (NLP, LLMs, Voice, Video generation) into containerized, low-latency production applications. Committed to engineering excellence, writing clean code, and building scalable software that delivers immediate business value.

CORE COMPETENCIES & TECHNICAL SKILLS

AI & ML Engineering:	Large Language Models (LLMs), vLLM, vLLM-Omni, Text-to-Speech (TTS), Voice Cloning, Video Generation Models, Prompt Engineering, RAG, Python, PyTorch, Scikit-learn, Transformers, CNNs, SpaCy, NLTK, Model Optimization.
Backend Serving & DevOps:	FastAPI (Asynchronous API Design), Node.js, Express.js, Docker, Docker Compose (Multi-container architecture), Ubuntu Linux, Neovim, Shell Scripting, Git, GitHub Actions.
Full Stack Web Development:	JavaScript (ES6+), TypeScript, React, Next.js, HTML5, CSS3, Tailwind CSS, PostgreSQL (SQL), MongoDB / Mongoose (NoSQL), Microservices.

PROFESSIONAL EXPERIENCE

Anivale — *AI Engineer Intern*

Feb 2026 – Oct 2026 | Remote, India

- **Multi-Modal AI Pipeline:** Architected and integrated a production-ready, end-to-end generative AI pipeline orchestrating advanced NLP/LLM text generation, low-latency voice cloning (TTS), and automated video generation models.
- **High-Performance Serving:** Implemented **vLLM** and **vLLM-Omni** to optimize LLM throughput and multi-modal inference execution, significantly lowering Time-to-First-Token (TTFT) and reducing operational costs.
- **Asynchronous API Design:** Built and optimized highly concurrent, asynchronous backend services using **FastAPI** to seamlessly stream data and manage workflows across separate AI model stages.
- **Containerized Deployment:** Standardized development, testing, and staging environments by containerizing the multi-model architecture using **Docker** and **Docker Compose**, ensuring seamless multi-container orchestration.
- **Infrastructure Management:** Configured and managed high-performance workloads within a native **Ubuntu Linux** environment, ensuring optimal GPU resource allocation and minimal system overhead.

EDUCATION & CERTIFICATIONS

Full Stack Open — *University of Helsinki* **Dec 2024**

Comprehensive, production-grade curriculum covering JavaScript, React, Redux, Node.js, Express.js, MongoDB, GraphQL, REST APIs, TypeScript, CI/CD pipelines, and Docker containers. Emphasized test-driven development and scalable architecture.

Independent Specializations

Deep structural study of Deep Learning architectures, computational linear algebra, and advanced calculus for machine learning systems.